Habib University

Course Title: Digital Logic and Design (EE-172 / CS-130 / CE-222)

Instructors:

Assignment No: 2 part A

Section:

Student Name:

Student ID:

Release Date: February 2nd, 2026

Total points: 55

Due by: February 12th, 2026

Points obtained:

Purpose:
The purpose of this assignment is to help you understand and apply the concepts and theorems of Boolean algebra for the implementation and simplification of logic functions. In addition, this assignment will also aid you to practice the tools and techniques used to minimize logic functions.

Instructions:
This assignment should be done individually.
All questions should be answered in **black ink only**.
Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:
Your assignments will be checked by instructor/TA.
You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.
Zero will be given if the assignment is found to be plagiarized.
Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:
1-day late submission – 20% deduction of the maximum allowable marks
2-days late submission – 40% deduction of the maximum allowable marks
No submission will be accepted after two days of the original deadline

Q#	Questions	Pts
1	Lamine Yamal is trying to send a secure tactical signal to his teammates during El Clásico to avoid an interception by the Madrid defense. The signal is sent as a sequence of footsteps encoded in binary. However, to prevent signal errors due to "lag" (glitches), the system must transmit in **Gray Code**. **a)** Design a logic circuit that converts an **8-bit Binary number** into Gray Code. **b)** Simulate the logic on **CircuitVerse** (recommended) or logic.ly and add the screenshot that shows the conversion of the binary number **10011101** to its equivalent Gray Code. **c)** Briefly explain why Gray Code is less prone to errors than Binary when values change rapidly (i.e., why does it stop the system from "crashing out"?).	4+4+2
2	Perform the following operations: **a)** Doctor Doom has cast a complex spell to shield his fortress. To break it, the Avengers need to find the **complement** of the logic shield. Find the complement of the expression below using De-Morgan's Law. Verify your solution by creating a truth table for the original and the complemented expression (use 4 variables x, y, z,w). $F = (x + y' + z)(x' + y + z')(w + x')$ **b)** Two "Brainrot" bots are communicating using 8-bit strings. Bot A sends $A = 10101100$ (The "Rizz" key) and Bot B sends $B = 01100110$ (The "Sigma" key). Evaluate the eight-bit result after the following bitwise logical operations: i. $A \text{ AND } B$ ii. $A \text{ OR } B$ iii. $A \text{ XOR } B$	4+3
3	Simplify the following Boolean expressions to the minimum number of literals to achieve "peak efficiency." Indicate the name of the theorem/postulate used at each step. **i.** $xy + x(y' + z) + y(y + z)$ **ii.** $(A + B)(A + C)(A' + B)$ **iii.** $w'xyz + wx'yz + wxy'z + wxyz'$	3x3
4	The **TVA (Time Variance Authority)** uses a pruning mechanism controlled by three "Timeline Monitors" (A, B, C). If a timeline is stable, the monitor reads **0**. If a timeline is branching/unstable, the monitor reads **1**.	2+3+3

	To conserve energy, the Pruning Stick (Z) only activates ($Z = 1$) if **exactly two** timelines are unstable at the same time. If zero, one, or all three are unstable, the stick remains off. **a)** Create a truth table for the Pruning Stick variable Z. **b)** Write the Minterms and Maxterms associated with the function. **c)** Express the Boolean function for Z in canonical SOP (Sum of Products) and Canonical POS (Product of Sums) forms.	
5	Using De Morgan's theorems, find the complement of the following functions. Do not simplify the result, just apply the complement to "mog" the original expression. **a)** $F(A, B, C, D) = (A' + B)(C' + D')(AB' + CD)$ **b)** $F(x, y, z) = x(y'z + x'y) + (x + z')'$	3x2
6	Identify the standard form (SOP or POS) of the following equations. Then, simplify the functions to the minimum number of literals. **a)** $F(A, B, C) = (A + B')(A + B' + C)(A + C)$ **b)** $F(X, Y, Z) = XY' + X(Y + Z') + Y(X' + Z)$	2x2
7	**VAR (Video Assistant Referee)** is checking a controversial goal. The logic system uses four sensors: Ball Crossed Line (W), Offside Flag (X), Foul Detected (Y), and Handball (Z). The decision function F is defined as: $F(W, X, Y, Z) = WX'Y'Z + W'X'Y'Z + WX'Y + WXY$ **a)** Obtain the truth table of F. **b)** Draw the logic diagram using the original (unsimplified) expression. **c)** Use Boolean algebra to simplify the function to a minimum number of literals (help the referee make a faster decision). **d)** Draw the logic diagram for the simplified expression and compare the number of gates required.	2+3+3 +3